Band-1/2 Computational Experiment: Union-Closed Sets — Gilmer Quantitative Remainder

Human + ZoraASI full-dash (executor)

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Band-1/2 Computational Experiment: Union-Closed Sets (Frankl) — Gilmer-Style Quantitative Remainder

Honesty stamp. This note does **not** claim that the Frankl union-closed sets conjecture is solved. Gilmer (2022) proved the first **positive constant** lower bound on element frequency. Subsequent work raised the constant to about 0.382. The **live Band-1–2 target** is strengthening that constant toward the conjectured $\frac{1}{2}$. We only finite-check the $\frac{1}{2}$ statement on small enumerated and random-closed families.

Abstract

We state Frankl's union-closed sets conjecture, ledger Gilmer's 0.01 breakthrough and the subsequent entropy-method improvements to $\frac{3-\sqrt{5}}{2} \approx 0.381966$ and mixture refinements ≈ 0.3823455, implement an exact finite-check pipeline on union-closed families, and identify the remaining gap ≈ 0.1177 (from ~ 0.382 to $\frac{1}{2}$) as the open quantitative remainder.

Box numbers: enumerated **5092** nonempty non-trivial union-closed families on $[n]$ for $n \leq 4$ — **all pass** frequency $\geq \frac{1}{2}$, with worst-case max-frequency exactly $\frac{1}{2}$; **55** classic constructions pass; **650** random union-closures on $n = 5..8$ pass. Runtime ≈ 0.120191 s.

(a) Original Frankl conjecture

Frankl's union-closed sets conjecture (1979). Let $\mathcal{F}$ be a finite family of sets that is **union-closed** ($A, B \in \mathcal{F} \Rightarrow A \cup B \in \mathcal{F}$). If $\mathcal{F} \neq \emptyset$ and $\mathcal{F} \neq \emptyset$, then there exists an element belonging to at least $\frac{1}{2}|\mathcal{F}|$ of the member sets.

Equivalently: in any nontrivial finite union-closed set system, some ground-set element has frequency $\geq \frac{1}{2}$.

(b) Gilmer 2022 breakthrough and subsequent constants (citations)

Bound	Approx.	Citation / note
First constant	0.01	Justin **Gilmer**, *A constant lower bound for the union-closed sets conjecture*, **arXiv:2211.09055** (Nov 2022). Entropy / information-theoretic method. Improves prior $\Omega(1/\log\lvert\mathcal{F}\rvert)$ bounds (Knill; Wójcik).
Gilmer-method optimum (i.i.d.)	$\frac{3-\sqrt{5}}{2} \approx 0.381966$	Same week: **Alweiss–Huang–Sellke** arXiv:2211.11731; **Chase–Lovett** arXiv:2211.11689; **Sawin** arXiv:2211.11504; **Pebody** arXiv:2211.13139.

Bound	Approx.	Citation / note
Mixture refinements	≈ 0.3823455	**Sawin** outline (convex combo i.i.d. + max-entropy coupling); evaluated by **Yu** arXiv:2212.00658 / *Entropy* 25(5):767 (2023) ≈ 0.38234; **Cambie** arXiv:2212.12500, $c \sim 0.3823455333667$.
Conditionally i.i.d. (ledger)	≈ 0.38271	**Liu** arXiv:2306.08824 — reported under numerically verified optimizer hypotheses; treat as conditional ledger, not a settled closed form.
Conjectured	$\frac{1}{2}$	Frankl — **still open**.

Caveat (path blocked): Gilmer's stronger information-theoretic conjecture that would have implied the full $\frac{1}{2}$ bound has **counterexamples** (Sawin; Ellis, independently). Entropy-method strengthenings continue, but the direct Gilmer-strengthening path to $\frac{1}{2}$ is closed.

Survey-style wrap: Cambie, *Progress on the union-closed conjecture and offsprings in winter 2022–2023*, arXiv:2306.12351.

(c) Finite exact-check pipeline (box)

Code: `/workspace/experiments/band1/union_closed/union_closed_probe.py`

Method. 1. Represent subsets of $[n]$ as bitmasks; a family is union-closed iff it is closed under bitwise OR. 2. **Full enumeration** for $n \le 4$: scan all $2^{2^n} - 1$ nonempty collections; retain those closed under OR; exclude the trivial $\emptyset$; compute element frequencies; require $\max_i f_i \ge \frac{1}{2}$. 3. **Classic constructions** for $n = 1..7$: power set, nonempty sets, principal filters, chains, singleton full set, singleton-generators closure, size-threshold closures. 4. **Random closures** for $n = 5..8$: sample random generators, take iterative union-closure, check frequency.

Enumeration ($n \le 4$) — all pass; sharp at $1/2$

n	# UC families (nonempty collections)	# checked (excl. $\emptyset$)	min of max-freq	all $\ge \frac{1}{2}$?	time (s)
1	3	2	**0.5**	True	3.2e-05
2	13	12	**0.5**	True	3.5e-05
3	121	120	**0.5**	True	0.000429
4	4959	4958	**0.5**	True	0.109006

Worst families attain **exactly** $\frac{1}{2}$ (sharp finite instances of the conjectured constant).

Random union-closures

n	trials	checked	min max-freq	mean max-freq	all $\ge \frac{1}{2}$?	time (s)
5	200	200	0.500000	0.8382	True	0.001647
6	200	200	0.571429	0.8583	True	0.002304
7	150	150	0.714286	0.8860	True	0.002258
8	100	100	0.750000	0.8977	True	0.002922

Classic constructions (sample of sharp / near-sharp)

name	n	$\lvert\mathcal{F}\rvert$	max-freq	pass
`power_set`	1	2	**0.5**	yes
`filter_contains_0_plus_empty`	1	2	**0.5**	yes
`total_chain`	1	2	**0.5**	yes
`power_set`	2	4	**0.5**	yes
`power_set`	3	8	**0.5**	yes
`power_set`	4	16	**0.5**	yes
`power_set`	5	32	**0.5**	yes
`power_set`	6	64	**0.5**	yes
`power_set`	7	128	**0.5**	yes

Totals: enum checked **5092** (pass=True); classic **55** (pass=True); random **650** (pass=True); wall **0.120191** s.

(d) Live Band-1–2 target — quantitative constant gap

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proven floor (unconditional Gilmer-method):   (3-√5)/2   0.381966
best mixture evaluation (Cambie/Yu):           0.3823455
Frankl conjecture:                            = 0.5
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absolute gap to close:                         0.1177
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Band-1–2 remaining work (not done tonight): new ideas beyond pure / mixture entropy couplings to push the constant from ~ 0.382 toward $\frac{1}{2}$, or prove the full conjecture by another route. Finite checks are consistent with Frankl but do **not** constitute a proof.

Reproducibility

- Code / results: `/workspace/experiments/band1/union_closed/` (`union_closed_probe.py`, `results.json`, `literature_ledger.json`)
- Log: `/workspace/experiments/logs/2026-09-11_union_closed.log`
- Seed for random closures: `20260911`

Conclusion

Literature: Gilmer $0.01 \to (3-\sqrt{5})/2 \approx 0.381966 \to$ mixture ≈ 0.3823455; Frankl $\frac{1}{2}$ **open**.
Finite box check: 5092 enumerated + 55 classic + 650 random families — **0 violations** of frequency $\geq \frac{1}{2}$; enumeration worst-case exactly $\frac{1}{2}$.
Live target: close the ≈ 0.1177 quantitative gap. **No claim that Frankl is solved.**